

Bounding the Answer Space — a worked 5b (Intro-to-ML)

*Guide §6a, step 5b. This is what “bound the answer space” actually produces, end to end, for one divergent assignment. Step 5a covers assessments with one right answer — a quiz item, a coding exercise — where the key falls out of the complete solution. This file is for the other kind: **many answers are valid, so there is no key to derive.***

1. The assignment (the divergent one)

Intro-to-ML’s convergent work is easy to assess: the notebook runs or it doesn’t. **CO5 is not like that.**

CO5 — Judge model fit using MSE/RMSE/ R^2 and **communicate results to stakeholders.**

The assignment: You’ve trained the Week 1 housing model. Write a **one-page model report** for a non-technical stakeholder who has to decide whether to use it. Say what it does, how well it works, and where it shouldn’t be trusted.

There is no single correct one-page report. Two good students will write genuinely different ones. That’s the definition of divergent work — and the reason step 2’s “build the complete solution first” doesn’t rescue you here.

2. The axes (write these first — they come from you, not the AI)

Before prompting anything, name the three or four things you’d comment on in **any** submission. These come from the annotated exemplar you built in step 2:

#	Axis	The question it answers
A1	Leads with the decision	Does the stakeholder know what to do by the end of the first paragraph?
A2	Honest about error	Is the model’s error expressed in units the reader cares about (dollars), not just R^2 ?

#	Axis	The question it answers
A3	Names the limits	Does it say where the model shouldn't be trusted — and is that specific, not hedging?
A4	No unexplained jargon	Can a reader who doesn't know what "residual" means still follow it?

Why first: if you let the AI generate the range before you've named the axes, you end up grading against whatever the model happened to vary.

3. The prompt (run it eight to ten times, varying the approach)

"You are a student in an introductory applied machine-learning course for working professionals. You have trained a linear regression model on the California Housing dataset (RMSE about \$50k, R^2 about 0.6). Write the one-page model report described below, for a non-technical stakeholder deciding whether to use the model.

[paste the assignment]

Write it as **one plausible student would** — not as a model answer. Vary your approach each time I ask: this run, be the kind of student who **[leads with the methodology / is over-confident about accuracy / buries the recommendation / writes well but avoids numbers / over-hedges everything]**."

Run it separately per variant. Asking once for "ten different versions" collapses them into one voice with ten hats on — you get variation in wording, not in *approach*, and approach is what you're trying to chart.

4. The feedback envelope (what came back, sorted by axis)

Axis	Strong	Typical	Off-base
A1 — leads with the decision	Opens with “I’d recommend using this for initial estimates, not final pricing.”	Recommendation appears, but in the last paragraph after the methodology.	No recommendation at all — describes the model and stops.
A2 — honest about error	“Typically within about \$50k of the real price — so it’s useful for ballparks, not for setting a listing price.”	Reports RMSE and R^2 correctly but never converts them into stakes.	“The model is 60% accurate.” (R^2 is not an accuracy percentage.)
A3 — names the limits	Names a specific boundary: the data is 1990s California; it won’t transfer to today’s market or another state.	Generic caution — “no model is perfect,” “more data would help.”	Claims the model is ready for production use.
A4 — jargon	Explains or avoids every technical term; a manager could read it cold.	One or two terms slip through unexplained (<i>residual</i> , <i>feature</i>).	Written for the instructor, not the stakeholder.

The most common failure was **A1** — burying the recommendation under the methodology — which is worth knowing *before* you teach the session, because it tells you what to emphasize in the segment.

5. The feedback bank (written before anyone submits)

Two or three sentences per band, in the course’s voice. Most of your marking is now done in advance:

A1 — leads with the decision

- *Strong*: “Your first sentence tells me what to do. That’s exactly right — a busy reader may not get past it, and they don’t need to.”
- *Typical*: “Your recommendation is sound, but it’s in the last paragraph. Move it to the top: the reader decides whether to keep reading in about ten seconds.”
- *Off-base*: “You’ve described the model accurately but never said what you’d do with it. A stakeholder can’t act on a description — what’s your

recommendation?”

A2 — honest about error

- *Strong*: “Converting RMSE into dollars is the move that makes this useful. That’s the sentence the reader will remember.”
- *Typical*: “The numbers are right, but what do they mean for someone making a decision? Try: ‘typically within about \$X of the real price.’”
- *Off-base*: “ R^2 isn’t an accuracy percentage — it’s the share of variance explained. Worth re-checking, because a stakeholder will take ‘60% accurate’ literally.”

A3 — names the limits

- *Strong*: “Naming the 1990s California data as the boundary is specific and honest. That’s the difference between a caveat and a disclaimer.”
- *Typical*: “‘No model is perfect’ is true of everything, so it tells the reader nothing. What specifically would make *this* model wrong?”
- *Off-base*: “This reads as production-ready. Given the error range, what would have to be true before someone relied on it?”

A4 — jargon

- *Strong*: “A manager could read this cold. That’s harder than it looks.”
- *Typical*: “Two terms slipped through — *residual* and *feature*. One clause each and you’re clear.”

6. Where this lives, and what students see

- **This file** (the envelope and the bank) → Instructor_Materials/reference_solution/solutions_and_key.m
It is for you, not for them.
- **The student-facing half** → a short reader’s checklist in Student_Materials/guided_practice/, phrased as questions they run against their own draft: *Does my first paragraph say what to do? Is my error in dollars? Have I named where this shouldn’t be trusted? Would my manager understand every word?*

Same four axes, two audiences. The checklist lets them self-correct before submitting, which is the point: **it’s a feedback envelope, not a grading envelope** — it tells you what to say, not what to score.

7. What this method can’t do

The limitations note in §6a applies, and it matters most here. The LLM is **not your students**: it writes like a competent median, so it will miss both the genuinely creative answer and the genuinely confused one. Treat the envelope

as a way to *widen* your expectations and pre-write your feedback — never as the definition of the acceptable set. If a student turns in something good that isn't in your table, the table was incomplete; that's normal.

Update it after the first run (§9) with what students actually submitted. The AI's version was a stand-in until you had real work to look at.